

LAB 1

REQUIREMENTS ENGINEERING & UML USE-CASE MODELLING

Problem Statement #09

FACULTY RESEARCH GRANT & PUBLICATION TRACKER

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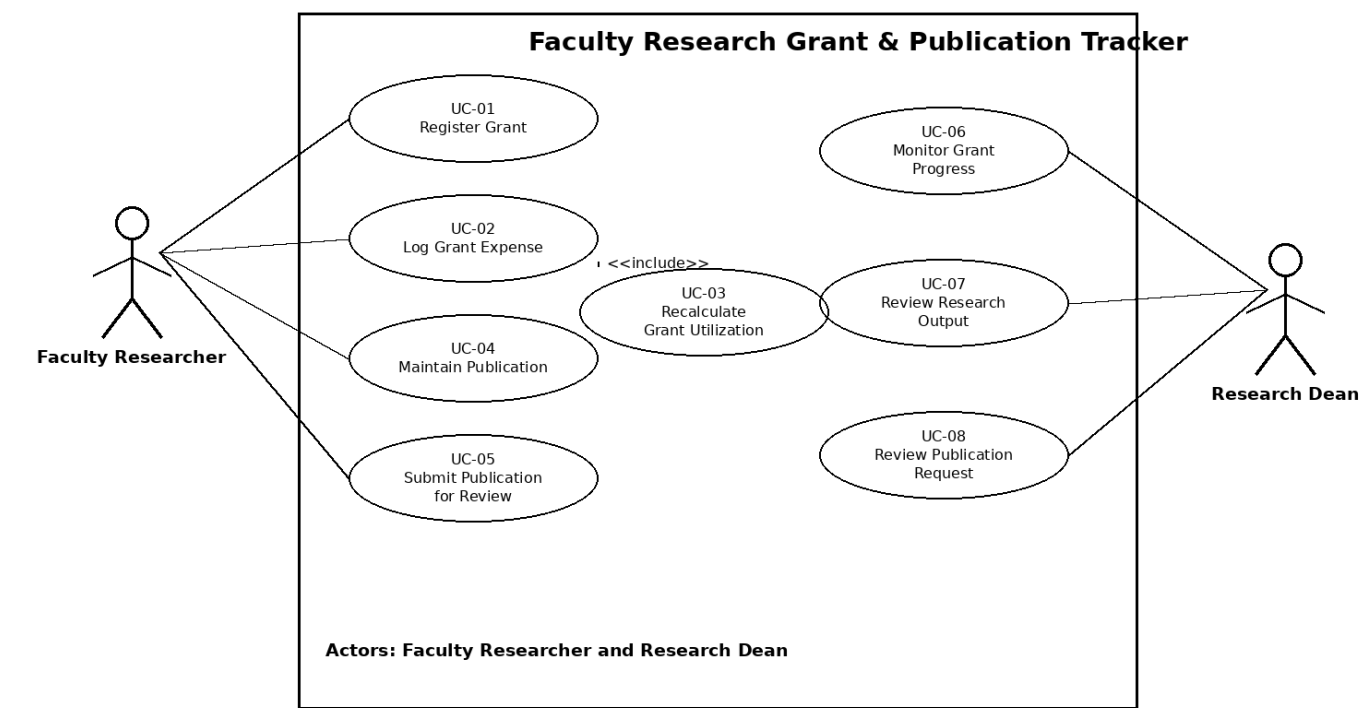
1. Complete Requirements Table

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall let Faculty Researchers register and maintain grant profiles containing sponsor, award period, approved ceiling, research milestones, and status.	High	Pass: a researcher can create or edit a grant and the saved profile shows the correct sponsor, dates, ceiling, and status. Fail: required grant details are lost, duplicated, or cannot be updated.	Creates one reliable record for the complete grant lifecycle and its planned milestones.
FR-002	Functional	The system shall let Faculty Researchers log research expenses by category, amount, date, grant, and receipt/reference number, while preventing spending beyond the available grant amount.	High	Pass: a valid expense is stored and the available amount decreases correctly; an over-limit expense is rejected. Fail: an invalid or excessive expense changes the grant balance.	Improves expense traceability and reduces accidental use of funds beyond the approved ceiling.
FR-003	Functional	The system shall maintain publication records with title, venue, authors, DOI or identifier, publication stage, indexing information, and citation count.	High	Pass: publication details can be added and later updated without losing existing information. Fail: publication metrics or identifiers are stored incorrectly.	Gives the research unit a consistent repository of scholarly output.
FR-004	Functional	The system shall allow a Faculty Researcher to submit a publication to co-authors for review and record comments, approval, or rejection status.	Medium	Pass: a submitted publication receives a visible review status and reviewer action. Fail: the request or its latest decision cannot be traced.	Provides a clear collaboration and review trail before a publication is finalized.
FR-005	Functional	The system shall provide the Research Dean with a dashboard showing grant utilization, upcoming milestones, active publications, and overall research-output indicators.	Medium	Pass: the dashboard reflects the current stored grant and publication data. Fail: displayed totals are missing, stale, or inconsistent with the records.	Helps the Research Dean identify delayed grants, high utilization, and research progress from one view.
NFR-001	Non-Functional	The system shall enforce role-based access so that Faculty Researchers can manage their permitted records while the Research Dean can view institutional summaries and review workflow information.	High	Pass: users only see actions allowed for their role. Fail: a user can perform an operation outside the assigned role.	Limits unauthorized changes and keeps responsibilities separated.
NFR-002	Non-Functional	The system shall keep a traceable activity history for grant, expense, publication, and approval	High	Pass: important changes show who performed them and when. Fail: a recorded change has no	Supports accountability when financial or publication records are reviewed later.

		changes, including the user and time of the action.		identifiable actor or timestamp.	
NFR-003	Non-Functional	The system shall remain responsive during normal concurrent use of grant entry, publication updates, approval actions, and dashboard viewing.	Medium	Pass: representative operations complete within an acceptable response time during simulated concurrent use. Fail: normal operations frequently time out or become unusable.	Keeps the portal practical when several researchers and administrators use it at the same time.

2. UML Use-Case Diagram

Actors: Faculty Researcher and Research Dean.



3. Use Cases

Use Case	Primary Actor	Purpose	Related Requirement(s)
UC-01 — Register Grant	Faculty Researcher	Create or update a grant profile with sponsor, award period, approved ceiling, milestones, and current status.	FR-001
UC-02 — Log Grant Expense	Faculty Researcher	Record a research expense against an active grant and attach its reference information.	FR-002
UC-03 — Recalculate Grant Utilization	Included behavior	Recompute total spending and the amount still available after a valid expense is recorded.	FR-002
UC-04 — Maintain Publication	Faculty Researcher	Add and update publication details, identifiers, indexing stage, and citation information.	FR-003
UC-05 — Submit Publication for Review	Faculty Researcher	Send a publication to the relevant co-authors and track review comments and decisions.	FR-004
UC-06 — Monitor Grant Progress	Research Dean	View grant utilization together with milestone progress and upcoming grant activities.	FR-005
UC-07 — Review Research Output	Research Dean	View publication activity, indexing information, and citation indicators across the research unit.	FR-003, FR-005
UC-08 — Review Publication Request	Research Dean	Review the status of publication collaboration requests and identify pending or completed decisions.	FR-004, FR-005

4. Core Use-Case Flow — Submit Publication for Review

Primary Actor	Faculty Researcher
Preconditions	The researcher is authenticated; the publication record exists; required author and publication details are available; the researcher is allowed to initiate a review request.
Postconditions	On success, the publication is marked as submitted for review and each selected co-author receives a pending review status. On failure, no review request is created and the publication remains in its previous state.
Main Success Scenario	1. Researcher signs in to the tracker. 2. System displays the researcher's publication records. 3. Researcher selects a publication that is ready for review. 4. Researcher chooses "Submit Publication for Review." 5. System displays the publication details and available co-authors. 6. Researcher selects the co-authors who should review the record. 7. Researcher optionally adds a review note. 8. Researcher submits the request. 9. System checks that required publication details are present. 10. System creates a review request for the selected co-authors. 11. System changes the publication workflow status to "Review Pending." 12. System records the submission time and submitting user. 13. System displays confirmation and the current review status.
Alternate Flow — Required Information Missing	1. Researcher submits the review request. 2. System detects that a required publication field or reviewer selection is missing. 3. System does not create the review request. 4. System identifies the missing information. 5. Researcher corrects the record and may submit again.
Alternate Flow — Reviewer Declines	1. A co-author opens the pending review request. 2. The co-author chooses to decline and enters a reason. 3. System records the decision and reason. 4. The publication status is updated to indicate that review action is required. 5. Researcher can revise the publication and submit a new review request.